

Camilo Artigas

(647) 836-0531 | cartigas@uwaterloo.ca

Dual US & Canadian Citizen

[Website](#) | [LinkedIn](#) | [GitHub](#)

Skills

ME: Siemens NX, SolidWorks, GD&T, DFMA, Structural/Modal FEA, FDM/SLA Printing, Mill, Lathe, CNC, Sheet Metal
SWE / FW: Python, C++ , Arduino, STM32, MATLAB, UART, RS-485/Modbus, Git, Ubuntu, RTOS, NumPy, Matplotlib
EE: Altium, LTSpice, Bring-up, Prototyping & Debugging, Soldering & Harnessing, Oscilloscope, Power Electronics, Sensors

Experience

Mechatronics Engineering Intern

Jan. 2026 – May 2026

Zipline *South San Francisco, CA*

- Designed full aircraft vibration fixture in NX for Busbar component testing, leveraging Modal Analysis to push first modal frequency beyond 500 Hz to enable launch-critical validation.
- Designed a Motor Rotor Tester in NX to validate magnet epoxy strength at 30,000+ RPM via a 1:5 belt drive reduction.
- Enabled full aircraft, droid & dock accelerated testing by retrofitting a shipping container into an environmental chamber, integrating an Arduino Opta and 8 RS-485 sensors to actuate heaters and misters via relay closed loop control.
- Developed multi-threaded C++ software for real-time control of chamber via RS-485 and a UART Python dashboard for data visualization; tuning duty cycles and tolerances to maintain 65 °C and 85% RH for 30 days.
- Integrated current sensors into existing 8-layer motor control PCB in Altium, owning schematics, layout, and bring-up while preserving full functionality and signal integrity.
- Owned an EV3 Battery test campaign by designing 2 PCBs in Altium and a 2000 Hz first-mode vibration fixture in NX.

Sensor Hardware Engineering Intern

May 2025 – Aug. 2025

Kodiak Robotics *Mountain View, CA*

- Designed and prototyped a liquid and air sensor cleaning system with 12 nozzles, optimizing pressure and angle for LIDAR, WFOV, and NFOV cameras — ensuring reliable perception for fully autonomous trucks.
- Designed and brought up a modular sensor cleaning test bench, integrating an in-house PCB with C++ firmware and Python test scripts to enable repeatable, plug-and-play validation across multiple sensors, reducing setup time by 70%.
- Designed a sheet metal enclosure in SolidWorks to house pneumatics for the air cleaning system, applying DFMA principles to reduce on-road troubleshooting time by 30% and personally mounting the full assembly onto the truck.
- Validated full cleaning system performance by WFOV/NFOV camera image capture and LiDAR PCAPs during operation.

Mechatronics Engineering Intern

Sep. 2024 – Dec. 2024

Sheartak Tools Ltd. *Waterloo, ON*

- 3D modelled 10+ spiral cutterheads in SolidWorks, drafting technical drawings using GD&T to specify critical tolerances.
- Developed Python scripts with OpenPyXL and NumPy, automating the migration of 500+ products, saving 30+ hours.
- Performed 6 installations on planers and jointers in less than 7 hours, installing 50+ carbide blades per cutterhead.

Student Design Teams

Hardware Director & Founder

Dec. 2025 – Present

Waterloo Automation Collective *Waterloo, ON*

- Leading all hardware and software development of UWaterloo's first injection molder. Running meetings, design reviews and mentoring 15+ students.
- Owning master-model of injection molder in SolidWorks, overseeing and modelling frame and injection unit assemblies.
- Leading in-house PCB development in Altium, prioritizing DFMA, cost optimization and performance.

Mechanical Lead & Designer

Aug. 2024 – Dec. 2025

Waterloo Aerial Robotics Group *Waterloo, ON*

- Co-led the mechanical team ahead of the 2026 competition, owning and managing the Task 1 Payload within the Hexacopter assembly. Leading meetings, design reviews and mentoring 15+ students.
- Led the design of the drone's landing gear in SolidWorks for the 2025 competition, angling its legs to ensure concentricity with bucket for water retrieval; implementing depth limiters, anti-tipping bars and carbon fiber members.

Education

University of Waterloo

Sep. 2023 – Apr. 2028

Candidate for a Bachelor of Applied Science in Mechatronics Engineering *Waterloo, ON*

Relevant Coursework: Sensors & Instrumentation, Mechanics of Materials, Embedded Systems, Materials Science

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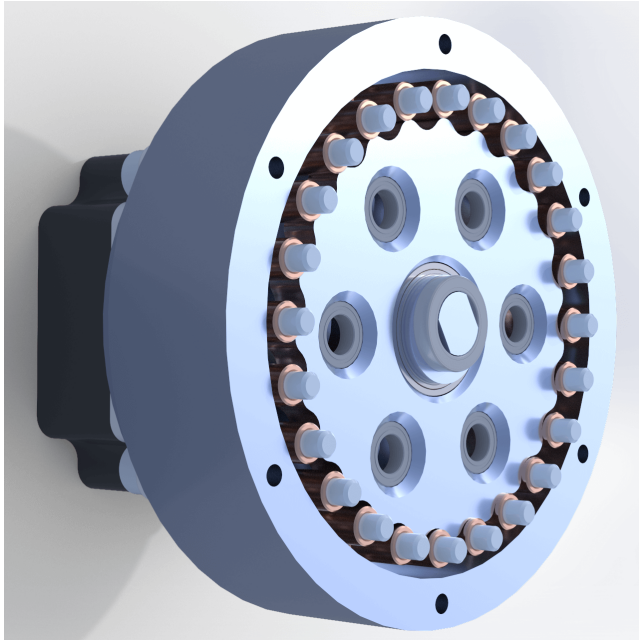
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Cycloidal Actuator | *SolidWorks, 3D Printing, DFMA, Python, Stepper Motors*

Designed a 130+ part assembly achieving a gear ratio of 23:1 using a cycloidal speed reducer. Developed a Python script for instant visualization of the cycloid, reducing design time by 50%. Successfully 3D printed a backlash-free prototype. ([Code](#))



SolidWorks Model



Functional Prototype → [WATCH HERE](#)

Sensor Cleaning @ Kodiak | *SolidWorks, Python, Systems Integration, Test Engineering, Pneumatics*

Designed and prototyped a liquid and air sensor cleaning system optimizing pressure and angle for a LiDAR and WFOV/NFOV cameras. Actuated it by bringing up a modular sensor cleaning test bench, integrating an in-house PCB with Python and C++ automation scripts. Validated full system by capturing synchronized images and LiDAR PCAPs during operation.



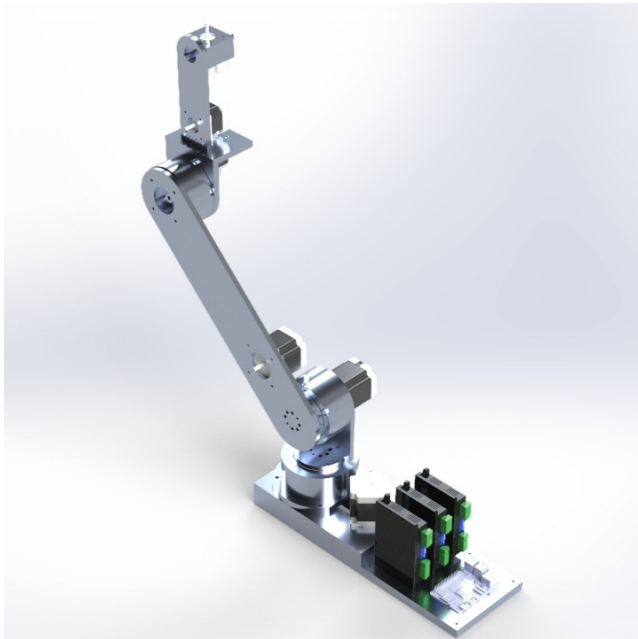
Kodiak Commercial Sensor Pod Assembly



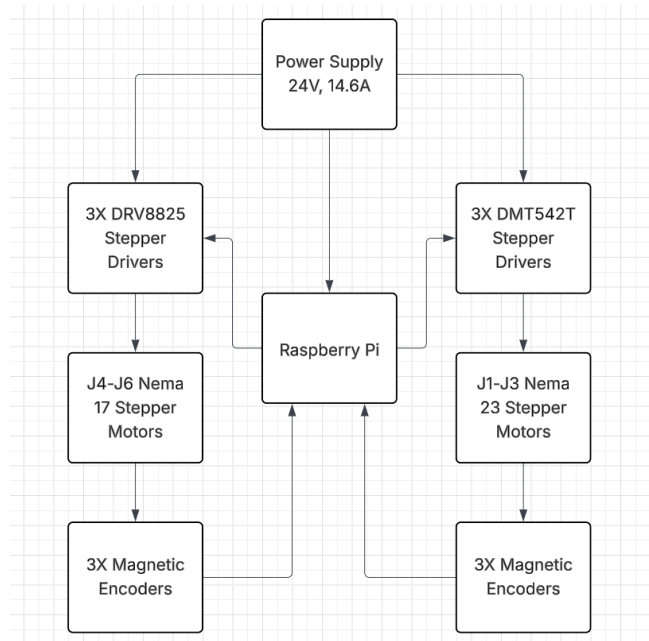
Sensors in cleaning system: WFOV, NFOV
Cameras and OT-128 LiDAR

6-DOF Robotic Arm | *SolidWorks, Altium, Power Electronics, ROS, PID Control, Systems Integration*

Designing a 6-DOF Robotic Arm capable of a 3 kg payload, integrating my own gearboxes, stepper motors, machined parts and inverse kinematics.



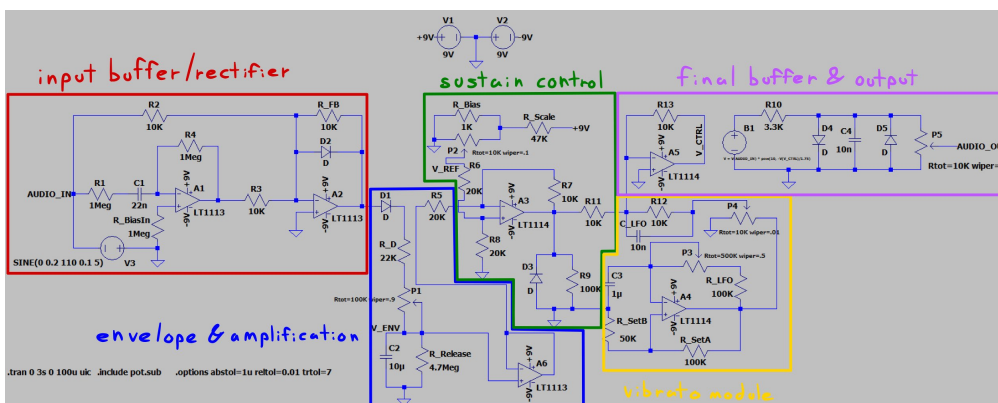
SolidWorks Model



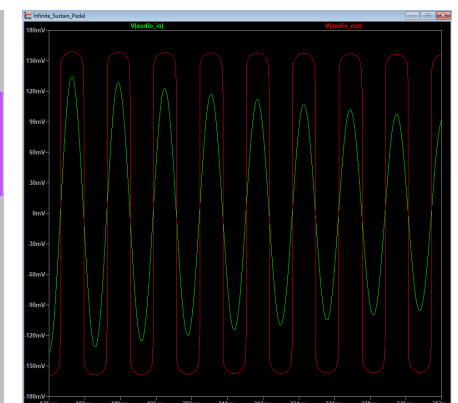
High Level Architecture

Infinite Sustain Guitar Pedal | *Altium, LT Spice, Signal Processing, Simulation*

Designed an infinite sustain guitar pedal, using 6 different stages of amplification and signal processing. Achieving infinite sustain (with threshold) for a plucked note. Currently working on layout in Altium and mechanical enclosure.



LT Spice Schematic



LT Spice Simulation
[Input → Green, Output → Red]

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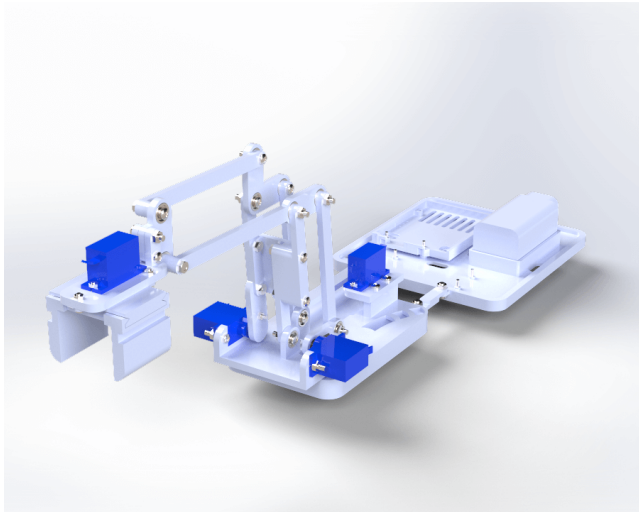
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3-DOF Robotic Arm | *SolidWorks, 3D Printing, DFMA, Python, C++ , UART, Servo Motors*

Designed and programmed a robotic arm by deriving its inverse kinematics, achieving smooth movement along the x, y, and z axes using servo motors. Established serial communication between a Python GUI and Arduino IDE for precise control. ([Code](#))



SolidWorks Model



Robotic Arm Controlled with GUI → [WATCH HERE](#)

Angled Landing Gear @ WARG | *SolidWorks, Test Engineering, Parametric Modelling*

Designed landing gear tailored towards WARG's competition challenge: water retrieval by landing on a barrel. Implemented parametric modelling, saving 10 hours. Added key features such as: optimal angle, anti-tipping bars and depth limiters. The landing gear was successful every time it was tested.



SolidWorks Overview



Landing Gear in Action → [WATCH HERE](#)